



GCE A LEVEL

1420QS

PHYSICS – Data Booklet

A clean copy of this booklet should be issued to candidates for their use during each Physics examination.

Centres are asked to issue this booklet to candidates at the start of the course to enable them to become familiar with its contents and layout.

Values and Conversions

Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
Fundamental electronic charge	$e = 1.60 \times 10^{-19} \text{ C}$
Mass of an electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Molar gas constant	$R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$
Acceleration due to gravity at sea level	$g = 9.81 \text{ m s}^{-2}$
Gravitational field strength at sea level	$g = 9.81 \text{ N kg}^{-1}$
Universal constant of gravitation	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
Speed of light in vacuo	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
Permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
Permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
Stefan constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Wien constant	$W = 2.90 \times 10^{-3} \text{ m K}$
Hubble constant	$H_0 = 2.20 \times 10^{-18} \text{ s}^{-1}$

$$T/\text{K} = \theta/^{\circ}\text{C} + 273.15$$

$$1 \text{ parsec} = 3.09 \times 10^{16} \text{ m}$$

$$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$$

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$

$$\frac{1}{4\pi\epsilon_0} \approx 9.0 \times 10^9 \text{ F}^{-1} \text{ m}$$

AS

$\rho = \frac{m}{V}$	$I = \frac{\Delta Q}{\Delta t}$																			
$v = u + at$	$I = nAve$																			
$x = \frac{1}{2}(u + v)t$	$R = \frac{V}{I}$																			
$x = ut + \frac{1}{2}at^2$	$P = IV = I^2R = \frac{V^2}{R}$																			
$v^2 = u^2 + 2ax$	$R = \frac{\rho l}{A}$																			
$\Sigma F = ma$	$V = E - Ir$																			
$p = mv$	$\frac{V}{V_{\text{total}}}\left[\text{or } \frac{V_{\text{OUT}}}{V_{\text{IN}}}\right] = \frac{R}{R_{\text{total}}}$																			
$W = Fx\cos\theta$	$T = \frac{1}{f}$																			
$\Delta E = mg\Delta h$	$c = f\lambda$																			
$E = \frac{1}{2}kx^2$	$\lambda = \frac{a\Delta y}{D}$																			
$E = \frac{1}{2}mv^2$	$d\sin\theta = n\lambda$																			
$Fx = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$	$n = \frac{c}{v}$																			
$P = \frac{W}{t} = \frac{\Delta E}{t}$	$n_1v_1 = n_2v_2$																			
efficiency = $\frac{\text{useful energy transfer}}{\text{total energy input}} \times 100\%$	$n_1\sin\theta_1 = n_2\sin\theta_2$																			
$F = kx$	$n_1\sin\theta_{\text{C}} = n_2$																			
$\sigma = \frac{F}{A}$	$E_{\text{kmax}} = hf - \phi$																			
$\varepsilon = \frac{\Delta l}{l}$	$p = \frac{h}{\lambda}$																			
$E = \frac{\sigma}{\varepsilon}$																				
$W = \frac{1}{2}Fx$																				
$\lambda_{\text{max}} = \frac{W}{T}$																				
$P = A\sigma T^4$																				
<table><tr><td></td><td colspan="2">leptons</td><td colspan="2">quarks</td></tr><tr><td>particle (symbol)</td><td>electron (e⁻)</td><td>electron neutrino (ν_e)</td><td>up (u)</td><td>down (d)</td></tr><tr><td>charge (e)</td><td>- 1</td><td>0</td><td>+ $\frac{2}{3}$</td><td>- $\frac{1}{3}$</td></tr><tr><td>lepton number</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>			leptons		quarks		particle (symbol)	electron (e ⁻)	electron neutrino (ν _e)	up (u)	down (d)	charge (e)	- 1	0	+ $\frac{2}{3}$	- $\frac{1}{3}$	lepton number	1	1	0
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A2

$\omega = \frac{\theta}{t}$	$C = \frac{\epsilon_0 A}{d}$
$v = \omega r$	$E = \frac{V}{d}$
$a = \omega^2 r$	$U = \frac{1}{2} QV$
$a = \frac{v^2}{r}$	$Q = Q_0 \left(1 - e^{-\frac{t}{RC}} \right)$ Capacitor Charging Charge + Voltage
$F = \frac{mv^2}{r}$	$Q = Q_0 e^{-\frac{t}{RC}}$
$F = m\omega^2 r$	$F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2}$ Colomb Law (Force)
$a = -\omega^2 x$	$F = G \frac{M_1 M_2}{r^2}$ $\frac{GMm}{r^2} = \frac{mv^2}{r}$
$x = A \cos(\omega t + \epsilon)$ $x_{\max} = A$ $v_{\max} = \omega A$ $a_{\max} = \omega^2 A$	$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$ Electric Field Strength $E = \frac{F}{q}$ $E = \frac{V}{d}$
$T = \frac{2\pi}{\omega}$	$g = \frac{GM}{r^2}$
$v = -A\omega \sin(\omega t + \epsilon)$	$V_E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$ Electric Potential $V = \frac{W}{q}$
$T = 2\pi \sqrt{\frac{m}{k}}$	$PE = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r}$ Electric Potential Energy
$T = 2\pi \sqrt{\frac{I}{g}}$	$V_g = -\frac{GM}{r}$
$pV = nRT$ and $pV = NkT$	$PE = -\frac{GM_1 M_2}{r}$ $V_{\text{esc}} = \sqrt{\frac{2GM}{R}}$
$p = \frac{1}{3} \rho c^2 = \frac{1}{3} \frac{N}{V} mc^2$ $Nm = M$	$W = q\Delta V_E$
$M / \text{kg} = \frac{M_r}{1000}$	$W = m\Delta V_g$
$n = \frac{\text{total mass}}{\text{molar mass}}$	$\frac{\Delta \lambda}{\lambda} = \frac{v}{c}$
$k = \frac{R}{N_A}$	$v = H_0 D$
$U = \frac{3}{2} nRT = \frac{3}{2} NkT$ $\frac{1}{2} mc^2 = \frac{3}{2} kT$	$\rho_c = \frac{3H_0^2}{8\pi G}$
$W = p\Delta V$	$r_1 = \frac{M_2}{M_1 + M_2} d$
$\Delta U = Q - W$	$T = 2\pi \sqrt{\frac{d^3}{G(M_1 + M_2)}}$
$Q = mc\Delta\theta$	$A = \lambda N$
$C = \frac{Q}{V}$	$N = N_0 e^{-\lambda t}$

$A = A_0 e^{-\lambda t}$	$F = Bqv \sin \theta$	Magnetic Force on Charge
$N = \frac{N_0}{2^x}$	$B = \frac{\mu_0 I}{2\pi a}$	Magnetic Force by two Wire
$A = \frac{A_0}{2^x}$	$B = \mu_0 nI$	Magnetic Force on Solenoid
$\lambda = \frac{\ln 2}{T_{\frac{1}{2}}}$	$\Phi = AB \cos \theta$	
$E = mc^2$	flux linkage = $N\Phi$	
$F = BIl \sin \theta$	Magnetic Force by Wire	

OPTION A

flux linkage = $BAN \cos \omega t$	$X_L = \omega L$
$V = \omega BAN \sin \omega t$	$X_C = \frac{1}{\omega C}$
$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$	$Z = \sqrt{X^2 + R^2}$
$V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$	$Q = \frac{V_L}{V_R} \left(= \frac{V_c}{V_R} \right)$
$V_{\text{rms}} = \frac{\omega BAN}{\sqrt{2}}$	$Q = \frac{\omega_0 L}{R}$

$$X = \frac{V}{I}$$

$$Z = \sqrt{(X_L - X_C)^2 + R^2}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

OPTION B

$I = I_0 e^{-\mu x}$	$f = 42.6 \times 10^6 B$
$Z = c\rho$	$H = DW_R$
$\frac{\Delta f}{f_0} = \frac{2v}{c} \cos \theta$	$E = HW_T$

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

OPTION C

$Ft = mv - mu$	$\tau = I\alpha$
$e = \frac{\text{Relative speed after collision}}{\text{Relative speed before collision}}$	$L = I\omega$
$e = \sqrt{\frac{h}{H}}$	$KE = \frac{1}{2}I\omega^2$
$I = \frac{2}{5}mr^2$	$p = p_0 - \frac{1}{2}\rho v^2$
$I = \frac{2}{3}mr^2$	$F_D = \frac{1}{2}\rho v^2 AC_D$
$\alpha = \frac{\omega_2 - \omega_1}{t}$	

OPTION D

$I = \frac{P}{A}$	$\frac{\Delta Q}{\Delta t} = -AK \frac{\Delta \theta}{\Delta x}$
$P = \frac{1}{2}A\rho v^3$	$P = UA\Delta\theta$

Mathematical Information

SI multipliers

Multiple	Prefix	Symbol
10^{-18}	atto	a
10^{-15}	femto	f
10^{-12}	pico	p
10^{-9}	nano	n
10^{-6}	micro	μ
10^{-3}	milli	m
10^{-2}	centi	c

Multiple	Prefix	Symbol
10^3	kilo	k
10^6	mega	M
10^9	giga	G
10^{12}	tera	T
10^{15}	peta	P
10^{18}	exa	E
10^{21}	zetta	Z

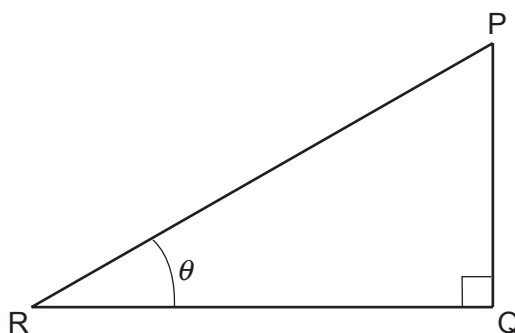
Areas and Volumes

$$\text{Area of a circle} = \pi r^2 = \frac{\pi d^2}{4}$$

$$\text{Area of a triangle} = \frac{1}{2} \text{ base} \times \text{height}$$

Solid	Surface area	Volume
rectangular block	$2(lh + hb + lb)$	lbh
cylinder	$2\pi r(r + h)$	$\pi r^2 h$
sphere	$4\pi r^2$	$\frac{4}{3} \pi r^3$

Trigonometry



$$\sin \theta = \frac{PQ}{PR}, \quad \cos \theta = \frac{QR}{PR}, \quad \tan \theta = \frac{PQ}{QR}, \quad \frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$PR^2 = PQ^2 + QR^2$$

Logarithms (A2 only)

[Unless otherwise specified 'log' can be $\log_e$ (i.e. $\ln$) or $\log_{10}$.]

$$\log(ab) = \log a + \log b$$

$$\log\left(\frac{a}{b}\right) = \log a - \log b$$

$$\log x^n = n \log x$$

$$\log_e e^{kx} = \ln e^{kx} = kx$$

$$\log_e 2 = \ln 2 = 0.693$$

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